# Kinds of municipal zone in Colombia: drinking-water risk, services and outcomes

## A descriptive cluster analysis, 2016–2023

Water-quality workshop project

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**Abstract** We cluster Colombian municipal zones (urban and rural) by utility-supplied drinking-water risk, service coverage and context with k-means, and describe how infant mortality, Saber 11 scores and school dropout differ between the resulting kinds. Outcomes are never used to form the clusters, and all comparisons are descriptive, not causal.

# 1 Introduction

Unsafe drinking water remains a public-health concern in rural Colombia. We ask which kinds of municipal zone exist in terms of water risk, services and context, and how health and education outcomes differ between them. Water risk is the utility-supplied ("prestadores") IRCA published by INS [1], scored and banded under Resolución 2115 de 2007 [2].

# 2 Methods

## 2.1 Data

The unit is a municipality's urban or rural zone in one year. Features come from INS [1], DNP TerriData [3], DANE [4]–[6], CHIRPS rainfall [7]–[9] and SRTM altitude [10]; outcomes from TerriData (infant mortality), ICFES [11] and MEN [12]. We keep 11,235 municipality zone-years with an IRCA value, 2016–2023.

## 2.2 Standardisation and clustering

Each feature $x_f$ is standardised within its year $t$,

$$z_{if} = \frac{x_{if} - \bar{x}_{f,t}}{s_{f,t}}, \tag{1}$$

and k-means [13] chooses centres $m_c$ that minimise the within-cluster sum of squares

$$W = \sum_{c=1}^{k} \sum_{i \in C_c} \|z_i - m_c\|^2. \tag{2}$$

We fit Equation 2 for $k = 2, \ldots, 8$ with 20 starts and choose $k$ by the elbow of $W$, checked with the mean silhouette [14]

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \tag{3}$$

where $a(i)$ is the mean distance from $i$ to its own cluster and $b(i)$ to the nearest other cluster. Stability across seeds is measured with the adjusted Rand index [15].

## 3 Results

The analysis chose $k = 4$ (silhouette 0.195; ARI across seeds $\geq 0.999$). Figure 1 shows what sets the kinds apart; Table 1 lists them.

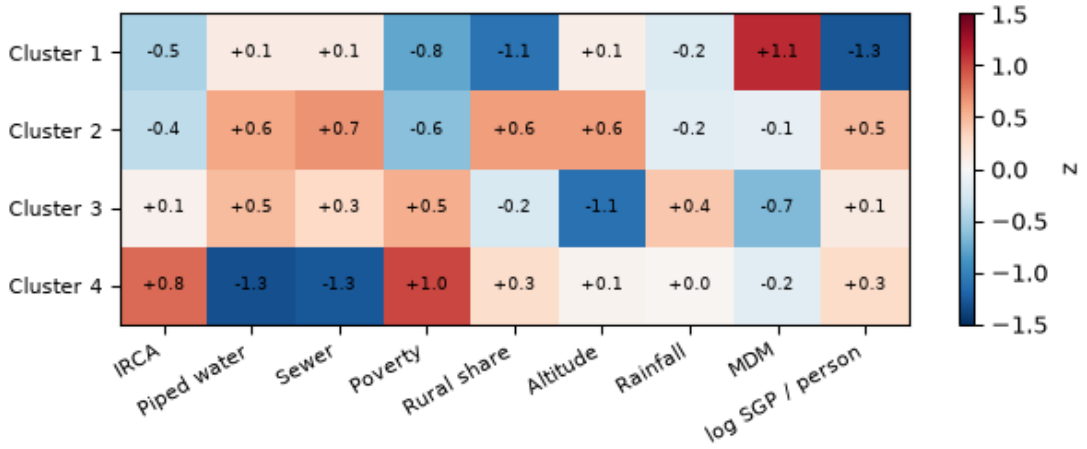



Figure 1: Cluster mean z-scores (Equation 1) by feature. IRCA is utility-supplied (prestadores).

Table 1: The kinds, with median utility-supplied (prestadores) IRCA and within-kind Spearman ρ between IRCA and infant mortality.

| cluster | kind | zone-years | median IRCA | ρ | n |
|---|---|---|---|---|---|
| 1 | Cities and large towns, no-risk water | 2271 | 1.6 | −0.04 | 1035 |
| 2 | Highland towns, no-risk water | 3750 | 4 | 0.1 | 1355 |
| 3 | Lowland towns, low-risk water | 2451 | 9.2 | 0.05 | 1482 |
| 4 | Rural, low-coverage, high-risk water | 2763 | 35.2 | 0.01 | 1290 |

# 4 Conclusions

The kinds differ far more in the features that formed them than in outcomes, and within each kind infant mortality hardly varies with IRCA band. These comparisons are descriptive: poverty, rurality, health-service access and reporting quality may confound them, and infant mortality is municipal while IRCA is zonal. They do not establish causal effects of water quality.

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